

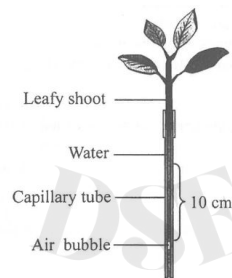
1. 2012Q3

Which of the following is the major cause that accounts for the ascent of water in trees?

- A. the loss of water from leaves
- B. the absorption of water in roots
- C. the thickness of tree trunks
- D. the diameter of xylem vessels

2. 2012Q10

Questions 10 and 11 refer to the following set-up, which is used to investigate the effect of different environmental conditions on the rate of water loss from a leafy shoot:

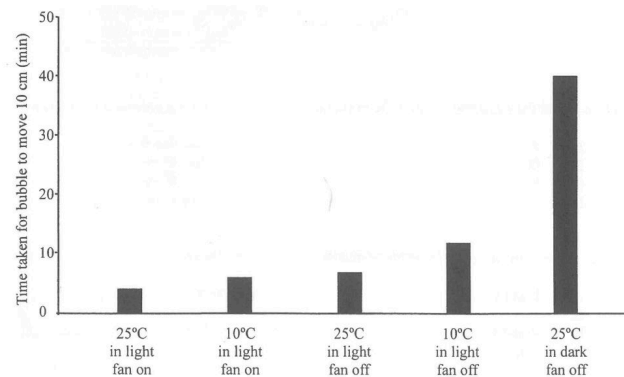


It is important to cut the leafy shoot under water at the beginning of the experiment because this

- A. allows the leafy shoot to adjust to the new environment.
- B. prevents the drying up of the cut end of the leafy shoot.
- C. avoids the forming of air bubbles in the xylem of the leafy shoot.
- D. makes sure that there is no water loss from the leafy shoot before the experiment.

3. 2012Q11

The results of the experiment conducted under different environmental conditions are shown in the bar chart below:



Which of the following changes in conditions results in the greatest decrease in water loss from the leafy shoot?

- A. switching off the light
- B. switching off the fan
- C. increasing the temperature from 10°C to 25°C
- D. decreasing the temperature from 25°C to 10°C

4. 2012Q12

Leaves were taken from four different plants and the number of stomata was counted. The results are shown in the following table:

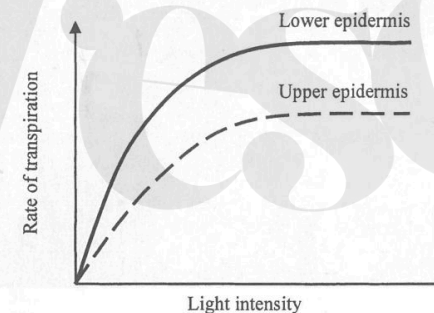
Plants	Mean number of stomata per cm ²	
	Upper surface of the leaf	Lower surface of the leaf
P	4000	28000
Q	0	800
R	8500	15000
S	8000	26000

Which plant will wilt first when it is grown in a very dry region?

- A. P
- B. Q
- C. R
- D. S

5. 2014Q20

Questions 20 and 21 refer to the graph below, which shows how the transpiration rates through the upper and lower epidermis of a leaf vary with light intensity:



Which of the following accounts for the difference in transpiration rates through the upper and lower epidermis shown above?

- A. The mesophyll layer near the lower epidermis has more air spaces.
- B. The upper epidermis is more exposed to light.
- C. The air temperature below the leaf is lower.
- D. The upper epidermis has fewer stomata.

6. 2014Q21

Which of the following methods can be used to measure the rate of transpiration through the upper and lower epidermis of the leaf?

- A. Peel the upper epidermis of the leaf and count the number of stomata under a microscope, and then repeat with the lower epidermis.
- B. Put the leaf into warm water and then count the number of bubbles that appear on each side of the leaf in a fixed period of time.
- C. Shine light on the upper epidermis of the leaf and measure the rate of water absorbed using a bubble potometer, and then repeat with the lower epidermis.
- D. Smear the upper epidermis of the leaf with vaseline and measure the rate of water loss using a weight potometer, and then repeat with the lower epidermis.

7. 2014Q22

The transpiration rate of a tree is much higher than that of a herbaceous plant because the tree

- A. is much taller than the herbaceous plant.
- B. has many more roots than the herbaceous plant.
- C. has many more leaves than the herbaceous plant.
- D. has much more xylem than the herbaceous plant.

8. 2014Q23

Which of the following cell types has the highest density of mitochondria?

- A. Root hair cells
- B. Leaf epidermal cells
- C. Spongy mesophyll cells
- D. Palisade mesophyll cells

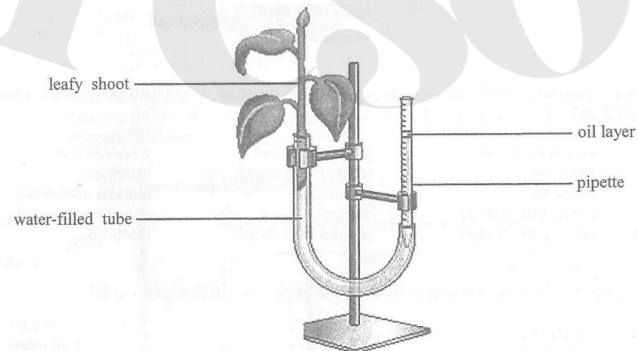
9. 2015Q2

Which of the following processes requires metabolic energy?

- A. glucose moves across the epithelium of the ileum
- B. carbon dioxide moves across the wall of air sacs
- C. oxygen moves into mesophyll cells
- D. water moves along the xylem

10. 2015Q17

Questions 17 and 18 refer to the set-up below, which is used to measure the rate of transpiration of a leafy shoot:



The assumption behind the use of this set-up for measuring the rate of transpiration is that

- A. the connections in the set-up are sealed off.
- B. the rate of water uptake is equal to that of water loss.
- C. the stomata of the leaves remain open throughout the experiment.
- D. the cutting of the shoot does not introduce air bubbles into the xylem vessels.

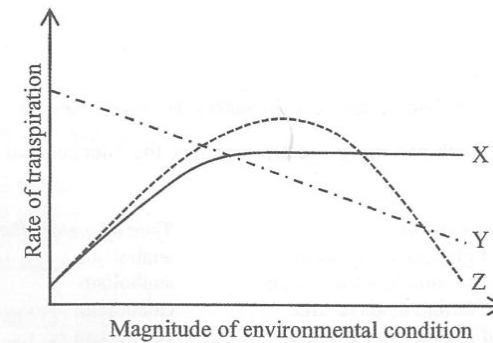
11. 2015Q18

Which of the following variables has the greatest influence on the rate of transpiration of the leafy shoot?

- A. the area of the leaves.
- B. the thickness of the leaves.
- C. the length of the leafy shoot.
- D. the number of xylem vessels.

12. 2016Q11

The graph below shows the changes of the transpiration rate of a plant under different environmental conditions:



Which of the following correctly matches the environmental conditions represented by X, Y and Z?

- | | X | Y | Z |
|----|-------------------|-------------------|-------------------|
| A. | light intensity | wind speed | relative humidity |
| B. | wind speed | light intensity | relative humidity |
| C. | relative humidity | light intensity | wind speed |
| D. | light intensity | relative humidity | wind speed |

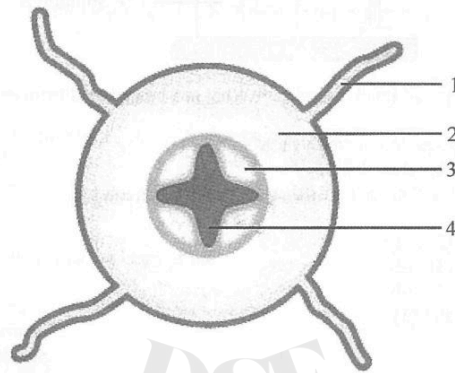
13. 2016Q12

A student wants to estimate the stomatal density of the upper and lower epidermis of a leaf using a microscope. Which of the following combinations correctly shows the magnification that should be used and the reason?

- | | Magnification | Reason |
|----|--------------------|-----------------------------|
| A. | low magnification | show a larger field of view |
| B. | low magnification | give a brighter image |
| C. | high magnification | show more cellular details |
| D. | high magnification | give a clearer image |

14. 2016Q21

Questions 21 to 23 refer to the diagram below, which shows the cross section of a young dicotyledonous root:



Which of the following parts provides rigidity to support the plant?

- A. 1
- B. 2
- C. 3
- D. 4

15. 2016Q22

Which of the following parts is responsible for the transport of food?

- A. 1
- B. 2
- C. 3
- D. 4

16. 2016Q23

Which of the following parts is most likely to have more mitochondria?

- A. 1
- B. 2
- C. 3
- D. 4

17. 2017Q13

Water absorbed by trees is mostly used

- A. to replenish water loss.
- B. for storage in vacuoles.
- C. as a raw material in photosynthesis.
- D. as a medium for reactions to take place.

18. 2017Q14

Questions 14 to 16 refer to the set-up below, which is used to investigate the effect of environmental factors on the transpiration rate of a leafy shoot. The leafy shoot was put into a beaker of red-coloured solution. After five hours, cross-sections of the shoot were cut starting from the top until red colour appeared in the cut section. The length of the remaining shoot was then measured.



Which of the following step(s) is / are necessary when preparing the set-up?

- (1) Smear vaseline onto the lower surface of the leaves.
 - (2) Add a layer of oil on the surface of the red-coloured solution.
 - (3) Cut the lower end of the leafy shoot under the red-coloured solution.
- A. (1) only
 - B. (3) only
 - C. (1) and (3) only
 - D. (2) and (3) only

19. 2017Q15

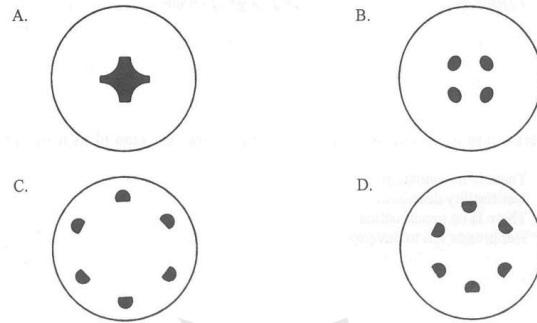
The length of the remaining shoot will be the longest if the experiment is conducted in

- A. hot and bright conditions.
- B. hot and humid conditions.
- C. cold and bright conditions.
- D. cold and humid conditions.

20. 2017Q16

A section of the remaining shoot was observed under a microscope. Which of the following diagrams correctly shows the appearance of the shoot section?

Key: ■ Stained red



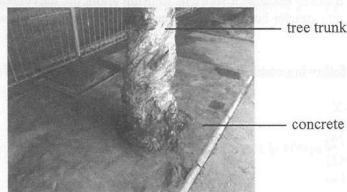
21. 2017Q17

Which of the following cell types contribute(s) to the support of a leaf in a woody flowering plant?

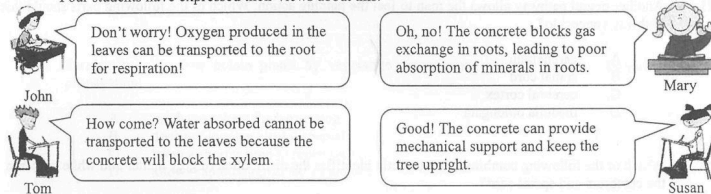
- (1) guard cells
- (2) xylem vessels
- (3) mesophyll cells
- A. (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

22. 2018Q24

The following photograph shows a tree with roots covered by concrete:



Four students have expressed their views about this:

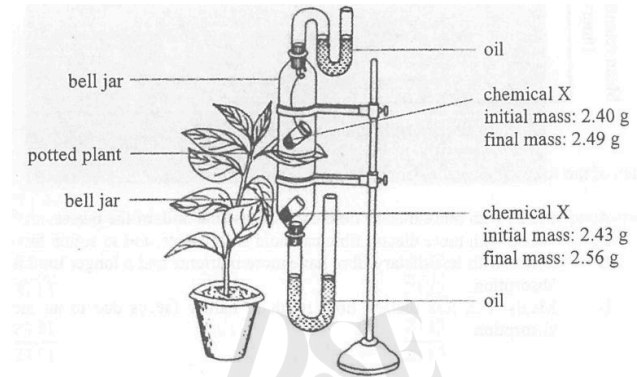


Whose view is correct ?

- A. John's view
- B. Mary's view
- C. Tom's view
- D. Susan's view

23. 2018Q30

Questions 30 and 31 refer to the set-up below. The set-up consists of two bell jars placed one above the other with the leaf of a potted plant in between. Chemical X was placed into the jars to absorb water vapour. The whole set-up was made air-tight. The masses of chemical X in the two jars were measured at the beginning and after five hours.



The change in mass of chemical X was mainly caused by

- A. water uptake by the plant.
- B. water loss in transpiration.
- C. water produced in respiration.
- D. water consumed by photosynthesis.

24. 2018Q31

Which of the following conclusions can be drawn from the results?

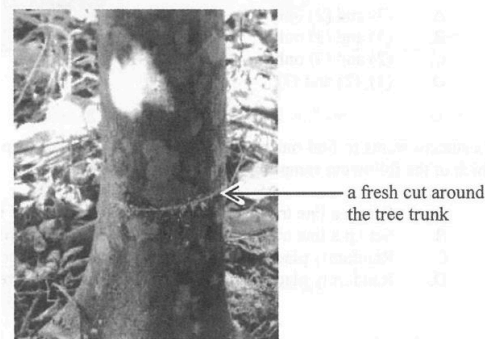
- A. Water absorption by the root is mainly driven by the lower epidermis of the leaf.
- B. There are more stomata at the lower epidermis than the upper epidermis of the leaf.
- C. The respiration rate is higher than the photosynthetic rate during the experiment.
- D. The photosynthetic rate of the upper layer of the leaf is higher than that of the lower layer.

25. 2018Q36

Which of the following processes mainly involves osmosis?

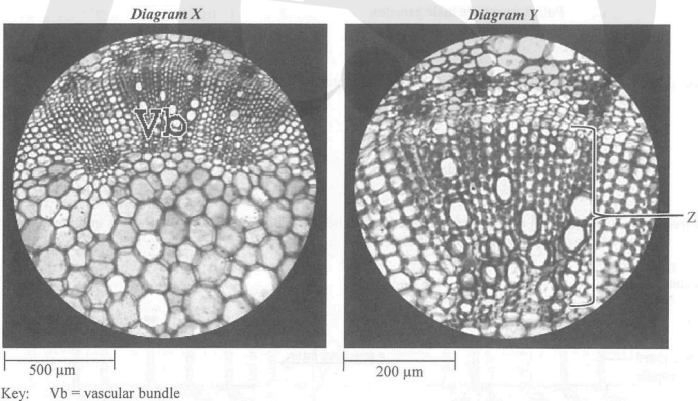
- A. movement of water along the xylem in plants.
- B. movement of water vapour out of stomata in plants.
- C. movement of water from tissue fluid to capillaries in humans.
- D. movement of water from tissue fluid to lymph capillaries in humans.

26. 2019Q32
The photograph below shows a fresh cut around the trunk near the bottom of a tree:



- The tree eventually died. Which of the following is the most likely reason for the death of the tree?
- A. Water could not be transported to the leaves for transpiration.
 - B. Water could not be transported to the leaves for photosynthesis.
 - C. Minerals could not be transported upward for protein synthesis.
 - D. Photosynthesis products could not be transported to the roots for respiration.

27. 2020Q22
Questions 22 to 24 refer to the photomicrographs below, which show the stem section of a plant observed under a microscope. Diagram Y shows a higher magnification of the vascular bundle (Vb) in Diagram X.



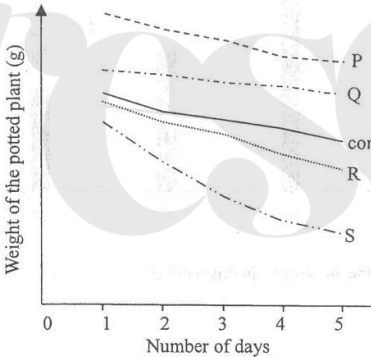
Which of the following combinations shows the correct steps for using the microscope in order to obtain the image in Diagram Y from that of Diagram X?

	Step 1	Step 2
A.	Move the slide towards the observer so that Vb is in the centre of the field of view.	Change the objective from 4X to 10X and adjust the focus.
B.	Change the objective from 10X to 40X and adjust the focus.	Move the slide towards the observer so that Vb is in the centre of the field of view
C.	Move the slide away from the observer so that Vb is in the centre of the field of view.	Change the objective from 4X to 10X and adjust the focus.
D.	Change the objective from 10X to 40X and adjust the focus.	Move the slide away from the observer so that Vb is in the centre of the field of view

28. 2020Q23
Which of the following descriptions of the function of tissue Z shown in Diagram Y is correct?
- A. It transports proteins upwards.
 - B. It transports minerals upwards.
 - C. It transports water downwards.
 - D. It transports sugars downwards.

29. 2020Q24
Which of the following structures is commonly found in all cells shown in Diagram Y?
- A. vacuole
 - B. cell wall
 - C. cytoplasm
 - D. cell membrane

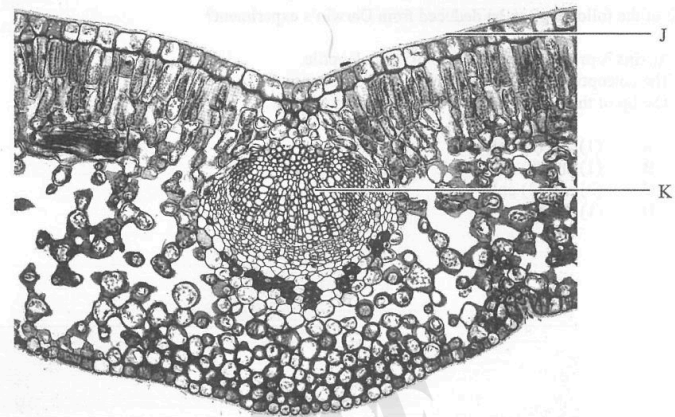
30. 2021Q30
Questions 30 and 31 refer to the graph below, which shows the effect of environmental conditions on the transpiration rate of a potted plant placed in a small room. The weight of the potted plant was recorded daily at noon for five consecutive days (control experiment). The experiment was repeated by changing one of the environmental conditions: increased light intensity, increased air current, increased relative humidity, or increased temperature.



- Which of the following lines represents the results with an increased relative humidity?
- A. P
 - B. Q
 - C. R
 - D. S

31. 2021Q31
In the above study, which of the following steps is necessary?
- A. Remove any fallen leaves.
 - B. Wrap the pot with a plastic bag.
 - C. Water the plant every day in the morning.
 - D. Smear vaseline on the lower epidermis of the leaves.

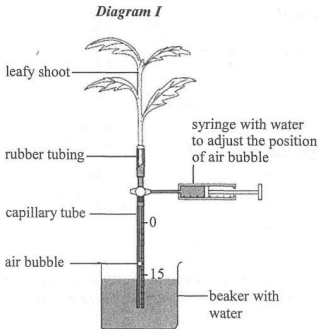
32. 2022Q29
Question 29 refers to the photomicrograph below, which shows a section of a leaf:



Which of the following substances will move out of the leaf through stomata from cell K in the daytime?

A. water
B. oxygen
C. mineral
D. carbon dioxide

33. 2022Q30
Questions 30 and 31 refer to the investigation below. Diagram I shows a set-up for measuring the transpiration rate of a leafy shoot under different environmental conditions. For each treatment, the experiment was conducted for three hours. Table II shows the initial and final readings of the position of air bubble in different treatments.



Treatment	Environmental conditions		Initial reading (cm)	Final reading (cm)
	Light intensity	Humidity		
1	Low	Low	14	7.4
2	Low	High	15	12.5
3	High	Low	15	5.6
4	High	High	14	10.6

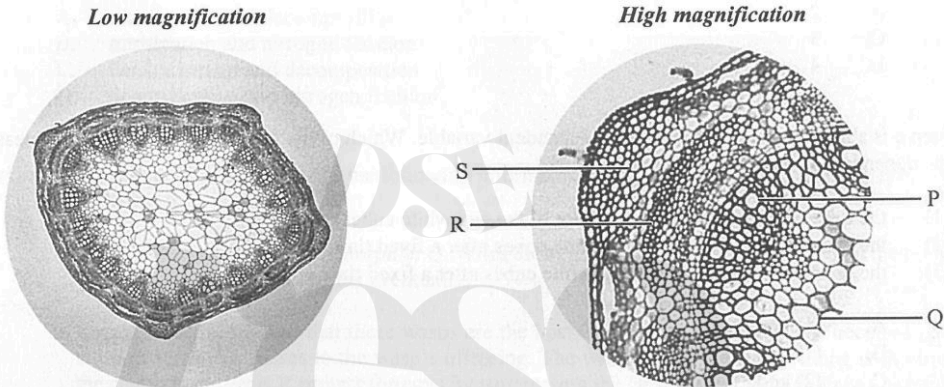
In which treatment does the leafy shoot have the highest transpiration rate?

A. 1
B. 2
C. 3
D. 4

34. 2022Q31
According to the results of the investigation, which of the following changes in environmental conditions will lead to a greater reduction in the transpiration rate of the leafy shoot?

A. At low humidity condition, adjust the light intensity from high to low.
B. At high humidity condition, adjust the light intensity from high to low.
C. At low light intensity, adjust the humidity from low to high.
D. At high light intensity, adjust the humidity from low to high.

35. 2023Q12
Questions 12 to 14 refer to the photomicrographs below, which show the images of the cross section of a certain part of plant viewed under different magnifications:



Which of the following combinations correctly matches the part of the plant where the section was taken from and the supporting evidence from the photomicrograph?

	<i>Gland</i>	<i>Enzyme</i>	<i>Optimum pH</i>
A.	gastric gland	carbohydrase	2
B.	liver	lipase	8
C.	salivary gland	amylase	11
D.	pancreas	protease	11

36. 2023Q13
The images above are shown to scale. Which of the following combinations correctly matches the eyepiece and objective used for the two magnifications (i.e eyepiece x objective)?

	<i>Low magnification</i>	<i>High magnification</i>
A.	10X x 4X	10X x 10X
B.	10X x 4X	10X x 25X
C.	10X x 4X	10X x 40X
D.	10X x 10X	10X x 40X

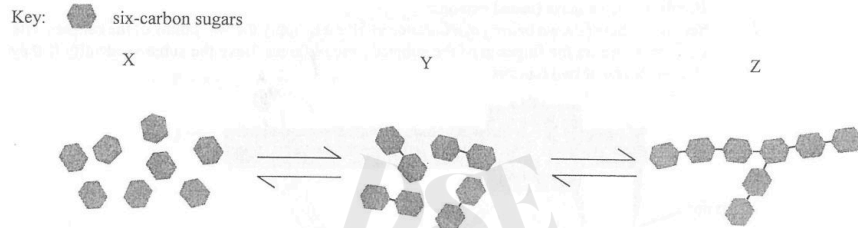
37. 2023Q14

Which of the labelled tissues is responsible for the transport of water?

- A. P
- B. Q
- C. R
- D. S

38. 2024Q19

Question 19 refers to the diagram below, which shows interconversion of three types of compounds catalysed by enzymes in a living organism:



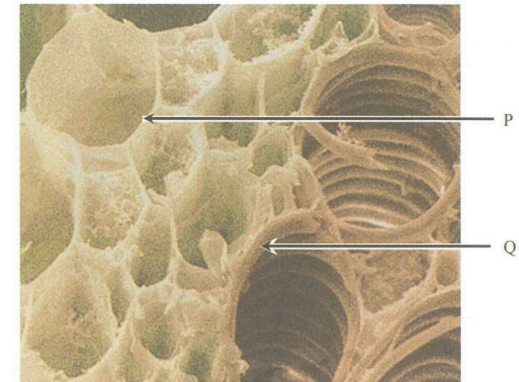
If the organism is a plant, which of the following statements is *incorrect*?

- A. X is the major form of carbohydrate transported through phloem.
- B. The conversion of Z to X takes place in a germinating seed.
- C. X can be the raw materials for synthesis of cell wall.
- D. The conversion of X to Z takes place in chloroplast.

LQ

1. 2012Q3

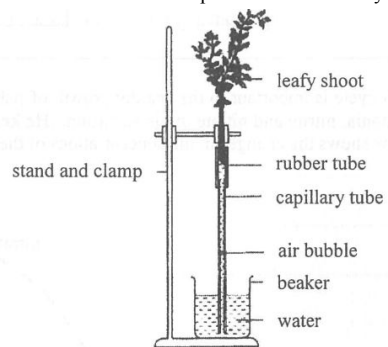
The electron micrograph below shows part of the stem of a plant containing two cell types, P and Q:



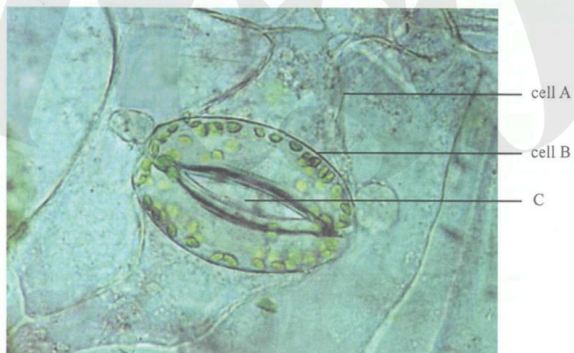
- a) Based on the photograph shown, state the difference between cell types P and Q in the structure indicated by the arrow heads. (1 mark)
- b) Describe how these cells contribute to the support of the plant. (4 marks)

2. 2013Q6

The following set-up can be used to determine the transpiration rate of a leafy shoot:



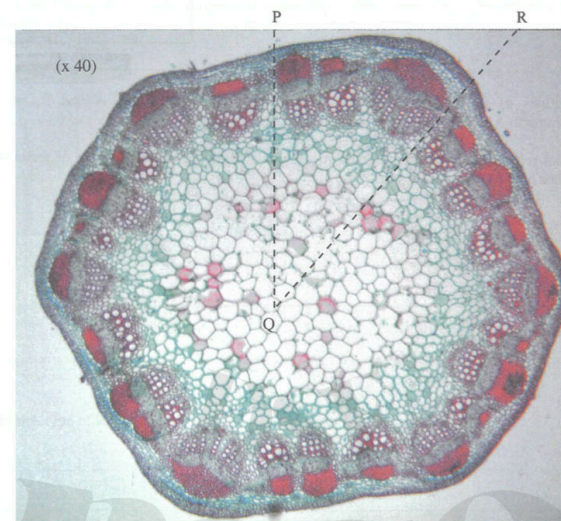
- In setting up this experiment, the lower end of the shoot should be cut under water. Why? (1 mark)
- Give **one** assumption for using this set-up to measure the transpiration rate. (1 mark)
- Explain how the transpiration rate will change if the fan placed near the shoot is switched on. (4 marks)
- The photomicrograph below shows the appearance of the surface of a leaf during daytime:



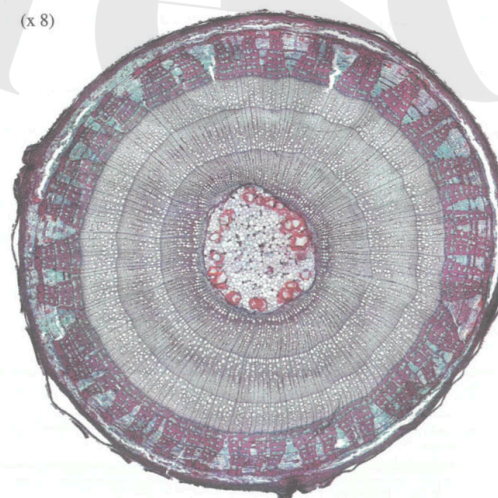
- In terms of sub-cellular structure, state **two** differences between cell A and cell B. (2 marks)
- What will happen to the size of C at night? Discuss the functional significance of this change. (3 marks)

3. 2014Q4

Photomicrograph A – Stem of plant A



Photomicrograph B – Stem of plant B



Cross sections of the stems from two different dicotyledonous plants, A and B, are shown in Photomicrograph A and Photomicrograph B.

- With reference to Photomicrograph A, draw a labelled low-power diagram of sector PQR in the space below. (5 marks)
- With reference to the photomicrograph, deduce the major means of support in plants A and B. (4 marks)

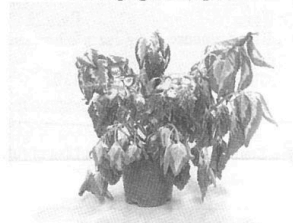
4. 2015Q9

The photographs below show the appearance of the leaves of a well-watered potted plant at 9 am and 1 pm on a sunny day in summer.

Photograph X (9 am)



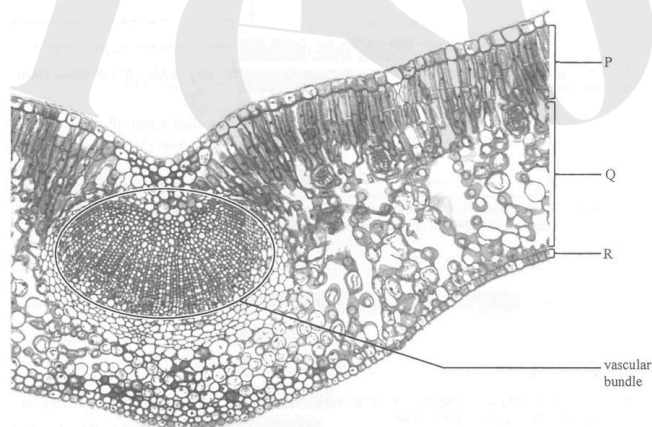
Photograph Y (1 pm)



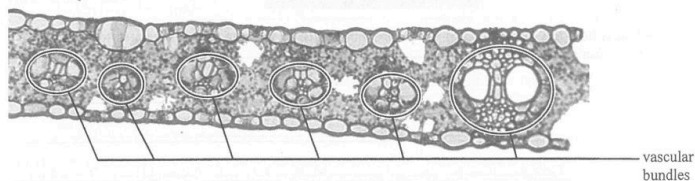
- Briefly describe how the appearance of the leaves shown in photograph X is maintained. (2 marks)
- Suggest an explanation for the appearance of the leaves at 1 pm (Photograph Y). (3 marks)
- With reference to the appearance of the leaves in the two photographs, which one is more effective for photosynthesis? Explain your answer. (4 marks)

5. 2018Q3

Photomicrograph X



Photomicrograph Y



Photomicrographs X and Y on the previous page show the cross sections of two leaves taken from different plant species.

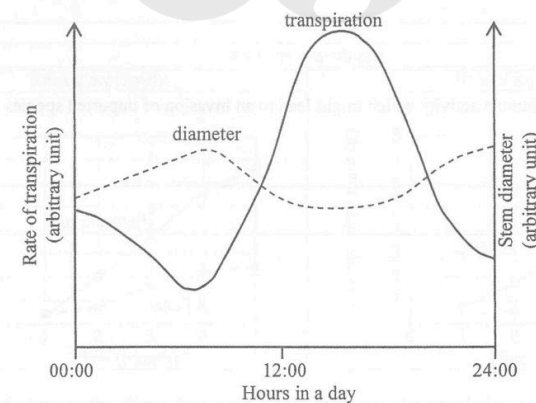
- Comparing the cell shapes of both leaves, which labelled tissue (P, Q or R) in Photomicrograph X is absent from Photomicrograph Y? (1 mark)
 - With reference to Photomicrograph X, what is the observable adaptive feature of the tissue identified in (i)? What is the significance of this adaptive feature? (2 marks)
- The leaf in Photomicrograph Y was taken from a plant species with leaves oriented vertically, as shown in the following paragraph:



Explain how the distribution of photosynthetic tissue in these leaves is related to the vertical orientation of the leaves. (3 marks)

6. 2019Q10

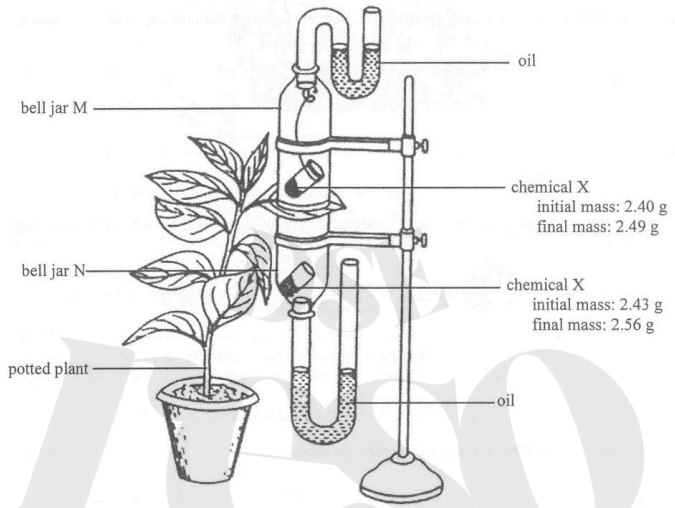
The graph below shows the change in the rate of transpiration and the change in stem diameter of a plant over 24 hours:



- Describe the relationship between the rate of transpiration and stem diameter. (1 mark)
- It is known that the change in stem diameter is related to the diameter of the xylem vessels. With reference to the way in which water is transported along the stem, explain the relationship between the rate of transpiration and stem diameter described in (a). (2 marks)
- Describe and explain **two** adaptive features of xylem vessels as a structure for water transport. (4 marks)

7. 2020Q5

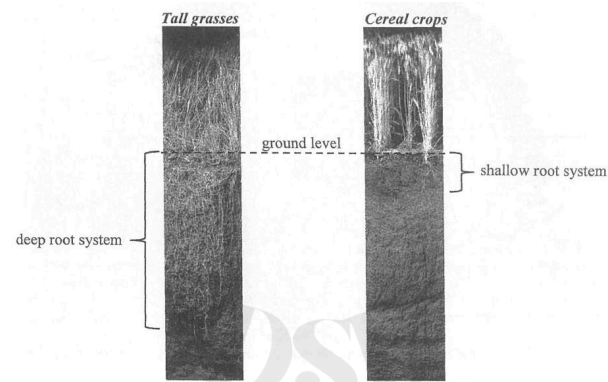
The diagram below shows an experimental set-up used to compare the transpiration rates of the upper and lower epidermis of the leaf of a potted plant. The set-up consists of two identical bell jars placed one above the other with the leaf of a potted plant in between. Chemical X was placed into the jars to absorb water vapour. The whole set-up was made air-tight. The masses of the chemical X in the jars were measured at the beginning and after five hours.



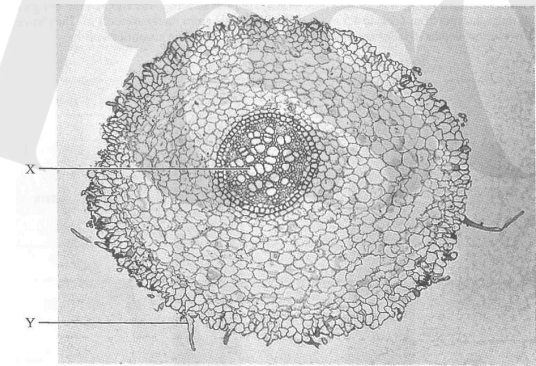
- a) i) Which of the following parameters is used to measure the transpiration rate of the plant in this investigation? Put a ‘✓’ in the appropriate box to indicate your choice. (1 mark)
- ☐ amount of water lost by the leaf
- ☐ amount of water absorbed by the leaf
- ii) How can the above set-up measure the parameter chosen in (a) (i)? (2 marks)
- b) List **two** variables which have been controlled by this set-up during the investigation. (2 marks)
- c) i) Compare the changes in the mass of chemical X in bell jars M and N. What conclusion can you draw? (2 marks)
- ii) Suggest **one** possible explanation for the conclusion. (1 mark)
- iii) Propose **one** workable method to test your hypothesis in (c) (ii). (2 marks)

8. 2022Q8b

Tall grasses and cereal crops belong to the same family and share a common ancestor. Cereal crops have been artificially selected for agriculture and their seeds harvested as food. The roots of tall grasses in grasslands range in depth from 1.5m to 4.5m while those of cereal crops rarely exceed 1m. The photographs below show the root depths of tall grasses and cereal crops under the same magnification:



b) The photomicrograph below shows the cross-section of the root of a cereal crop:



i) Complete the table below to show how an observable feature of the structures X and Y is related to its function. (4 marks)

	Oversable feature	Function
X		
Y		

ii) Explain how water is transported from structure Y to structure X. (3 marks)

9. 2023Q2b

All cells are derived from stem cells. They undergo differentiation in which the cells change in form and shape which enable them to perform specialized functions.

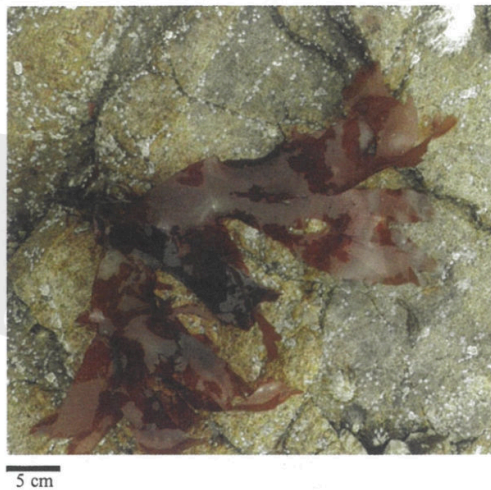
- b) Suggest a type of plant cell which also experiences degradation of cellular components during differentiation. Explain the significance of the degradation to the function of the cell type. (2 marks)

10. 2023Q5d

- d) Describe how the photosynthetic products of the leaves are stored in the underground tubers of a potato plant. (3 marks)

11. 2024Q7b, c, d

- b) In a field trip to a rocky shore, a student found a thin brown sheet lying on a rock. The student suspected that it was an alga. A small piece of the sample was taken back to school for further study.

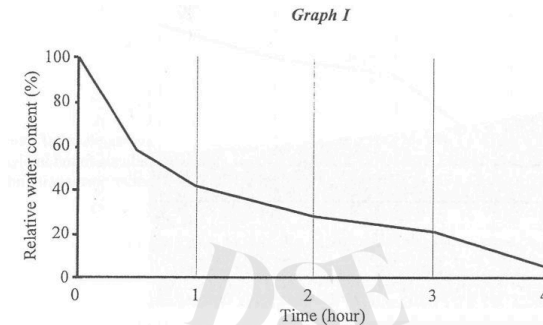


Using apparatus and reagents available in a school laboratory, design a set-up which can be used to show whether the sample of this thin sheet can undergo respiration. In the space below, draw a simple labelled diagram of this set-up. (3 marks)

Title: A set-up for demonstrating if respiration occurs in the sample

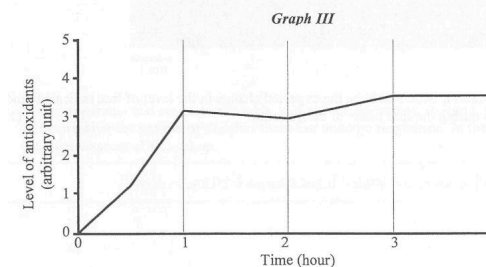
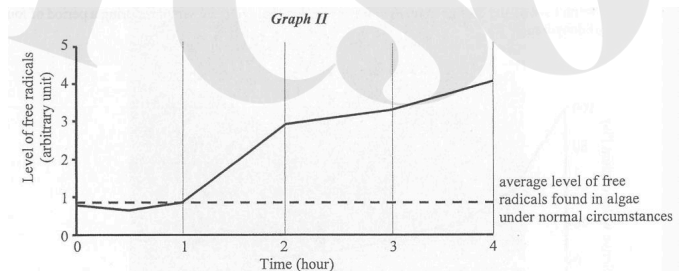
- c) Under normal circumstances, the level of free radicals in algal tissues is kept at a certain level as a result of homeostasis. In response to dehydration, algal tissues will be stimulated to produce free radicals which can cause damage to the cell components if there is an accumulation.

Graph 1 shows the change in relative water content of algal tissue samples during a period of four hours of dehydration.



With reference to Graph I, what would be the expected change in the level of free radicals found in the algal tissue samples during the four hours of dehydration? (1 mark)

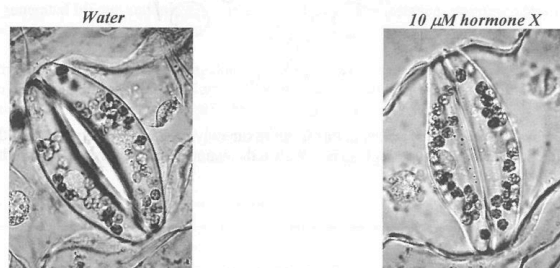
- d) Graph II and Graph III respectively show the actual change in the level of free radicals and the change in the level of antioxidants found in the algal tissue samples of the alga during the same period of dehydration:



Based on your answer in (c) and the data shown in Graph II and Graph III, suggest the role of antioxidants in helping the algae to cope with the dehydration. Give *two* pieces of evidence from the data shown. (3 marks)

12. 2024Q9

Hormone X is a plant hormone which is produced in leaves of plant P when water supply is inadequate. A student detached some leaves from plant P and placed them in either water or a $10\mu\text{M}$ solution of hormone X. After two hours, the student examined the lower epidermis of the leaves under a light microscope. The photomicrographs below show the images obtained:



- a) Based on the above information, explain the importance of hormone X to the drought tolerance in plant P. (2 marks)
- b) In nature, there are varieties of plant P which produce different amounts of plant hormone X in response to drought stress. The student measured the fresh leaf masses of two different varieties (A and B) of plant P after drought treatment for two weeks. The results are shown in the table below:

Plant variety	Treatment	Leaf fresh mass (g)
A	Control	0.20
	Drought	0.18
B	Control	0.21
	Drought	0.08

Which variety will have a higher level of hormone X produced? Explain your answer. (3 marks)

Ans

2012Q3: A (68%)	2012Q10: C (69%)	2012Q11: A (64%)	2012Q12: D (48%)
2014Q20: D (78%)	2014Q21: D (65%)	2014Q22: C (73%)	2014Q23: A (61%)
2015Q2: A (78%)	2015Q17: B (75%)	2015Q18: A (78%)	2016Q11: D (73%)
2016Q12: A (69%)	2016Q21: D (71%)	2016Q22: C (85%)	2016Q23: A (73%)
2017Q13: A (57%)	2017Q14: B (53%)	2017Q15: A (76%)	2017Q16: D (61%)
2017Q17: C (51%)	2018Q24: B (51%)	2018Q30: B (76%)	2018Q31: B (81%)
2018Q36: C (66%)	2019Q32: D (50%)	2020Q22: C (28%)	2020Q23: B (64%)
2020Q24: B (63%)	2021Q30: B (44%)	2021Q31: B (30%)	2022Q29: A (61%)
2022Q30: C (91%)	2022Q31: D (33%)	2023Q12: D (69%)	2023Q13: D (21%)
2023Q14: A (66%)	2024Q19: A (39%)		

2012Q3

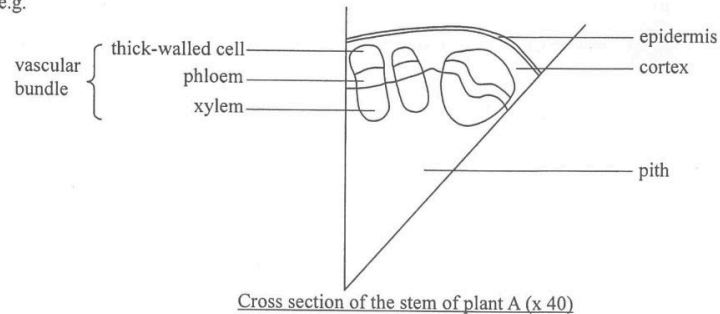
- 3 (a) • the cell wall of cell type Q is much thicker than that of cell type P (1)
- (b) • when there is ample supply of water (1)
- cell type P provides turgidity to the plant (1)
- cell type Q has thickened cell wall (1)
- which provides rigidity to the plant (1)
- 5 marks

2013Q6

- 6 (a) • avoid blockage of xylem by the air bubble formed during cutting (1)
- (b) • The rate of transpiration is the same as the rate of water absorption / the water absorbed is used for transpiration only (1)
- (c) • the transpiration rate will increase (1)
- Because the air current sweeps away the water vapour around the leafy shoot (1)
- The concentration gradient of water vapour between the atmosphere and the air space in the leaves becomes steeper (1)
- water vapour diffuse out to the atmosphere at a faster rate (1)
- (d) (i) • cell B contains chloroplasts but cell A does not (1)
- the cell wall of cell B has uneven thickness while that of cell A is even (1)
- (ii) • the size of C will be reduced (1)
- to reduce transpiration / water loss (1)
- when the need for gas exchange decreases in the absence of photosynthesis (1)
- 11 marks

2014Q4

4. (a) Title (1)
Resemblance of drawing (1)
Labels, any three (1, 1, 1)
e.g.



- 4 (b) • the stem of plant A has a large proportion of thin-walled cells (1),
• therefore plant A is mainly supported by turgidity of the thin-walled cells (1)
• While the stem of plant B has a large proportion of xylem / thick-walled cells / woody tissues (1)
• therefore it is mainly supported by mechanical or physical strength / rigidity of the thick-walled cells / xylem / woody tissues (1)

2015Q9

- 9 (a) • it is maintained by the rigidity of the veins / vascular bundles / xylem running through the leaves (1)
• and the turgidity of the mesophyll cells / thin-walled cells (1)
- (b) • a large amount of water is lost from the leaves by transpiration (1) / transpiration rate is very high (1)
• water absorption rate cannot keep up with the rate of water loss (1)
• mesophyll cells lose water and become flaccid (1)
thus can no longer support the leaves to maintain their upright position / so the leaves become drooped
- (c) • appearance shown in photograph X (1)
• the leaves are flat / fully extended (1)
• and thus the exposed surface can be oriented towards the light source (1)
• therefore maximizing light absorption for photosynthesis (1)

9 marks

2018Q3

- 3 (a) (i) • P (1) (1)
- (ii) • closely packed / located at the upper side facing the sun directly (1)
• this maximises light absorption for photosynthesis (1) (1+1)
- (b) • when the leaves are oriented vertically, both sides of the leaves have chance of receiving sunlight (1)
• therefore, the photosynthetic tissues are evenly distributed on both sides of the leaf (1)
• such that photosynthesis can be carried out effectively regardless of the change in the orientation of sun / the direction of illumination from sun during daytime (1)

6 marks

2019Q10

- 10 (a) • the diameter of the stem decreases as the rate of transpiration increases / the diameter of the stem increases as the rate of transpiration decreases (1) (1)
- (b) • water is transported up the stem by transpiration pull / pulling force / suction force / negative pressure (1)
• this force increases with transpiration rate and pulls the xylem vessel's walls inwards (1)
thus reducing the diameter of the xylem as well as that of the stem (1) (2)
- (c) • hollow tube (1) to allow the passage of water with low resistance (1)
• thickened / lignified wall (1) to withstand the negative pressure of the transpiration pull / to prevent the collapse of xylem vessels (1) (4)

7 marks

2020Q5

- 5 (a) (i) amount of water lost by the leaf (1) (1)
- (ii) • the water vapour is lost from the leaf surface through transpiration (1)
• the mass of chemical X increases after absorbing water vapour (1) indicating the mass of water lost from the epidermis (2) (2)
- (b) Any **two** of the following:
• type of plant or leaf (same potted plant / same leaf being investigated) (1)
• surface area of the epidermis (covered by the identical flasks) (1) (2)
• wind speed (enclosed system) (1)
• Volume of air containing the water vapour (confined by the identical flasks) (1)
- (c) (i) • the increase in the mass of chemical X in bell jar N is greater than that in bell jar M (1)
• this shows that the rate of water lost from the lower epidermis is higher than that of the upper epidermis (1) (2)
- (ii) this is probably due to: (any **one** of the following)
• different number of stomata in the upper and lower epidermises (1)
• different stomatal density of the upper and lower epidermises (1)
• different stomatal structures of the upper and lower epidermises (1)
• different thickness of the cuticles of the upper and lower epidermises (1)
- (iii) number of stomata / stomatal density: (any **one** set)
• microscopic examination of epidermal peel (1), count the number of stomata per unit area in the upper and lower epidermis (1)
• use nail polish to obtain a mold of the epidermis for microscopic examination (1), count the number of stomata per unit area in the upper and lower epidermises (1)
• immersion of the leaf in hot water (1), count the number of bubbles appearing on each surface per unit area of the upper and lower epidermises (1) (2)
- OR
- different stomatal structure / thickness of cuticle:
• cut a section of the leaf for microscopic examination (1), compare the appearance of the stomata / the thickness of the cuticles of the upper and lower epidermises (1)

10 marks

2022Q8b

- 8 (b) (i)
- | | Observable feature | Function |
|---|--------------------------------|---|
| X | large empty lumen / hollow (1) | offers less resistance to water transport (1) |
| | thick cell wall (1) | prevent collapse of the vessel (1) |
| Y | slender / elongated (1) | penetrates the space between soil particles / increases surface area for water / mineral absorption (1) |
- (4)
- (ii) • transpiration pull continues to draw water from cells adjacent to X (1), lowering the water potential of neighbouring cells (1)
• absorption of water at Y maintains a high water potential there (1)
• Water moves from Y to x along the water potential gradient across the root (1) (max. 3)

2023Q2b

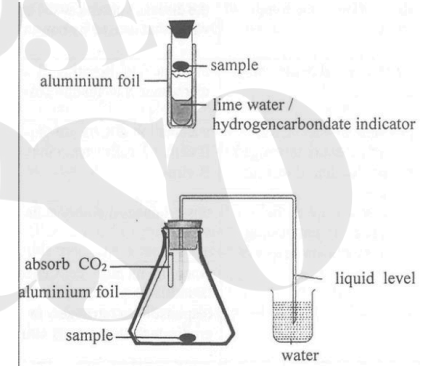
- 2 (b) Any **one** of the following
• xylem tube / xylem vessel (1) with all the cell content degraded to form a hollow tube which provides little resistance to water transport (1)
• sieve tube (1) with reduced cell content which offers a lower resistance to the translocation of sucrose (1) (2)

2023Q5d

- 5 (d) Concept for mark award:
• conversion of photosynthetic products to transport form (1)
• path of transport including tissue involved (1)
• conversion of transport form to storage form (1) (3)
- e.g.
• photosynthetic products (e.g. 3-C compounds, glucose, triose phosphate, starch, carbohydrates) are converted to sucrose for transport (1)
• along the phloem from leaves through stem, and then to the underground tubers (1)
• sucrose is converted to starch for storage in the underground tubers (1)

8 marks

2024Q7b, c, d

	Concept for mark award	Example	Marks
7(b)	the set-up should show • a closed system / an air pump system • blockage of light • valid method to show the change in carbon dioxide		(3)
7(c)	prediction based on the information	the level of free radicals should increase continuously from 0 to 4 hours	(1)
7(d)	inhibitory action of antioxidants	antioxidants counteract the action of free radicals / reduce the level of free radicals	(1)
	supporting evidence from the graph	from 0 to 1 hour, the level of antioxidants increases to suppress the level of free radicals in the algal tissue from 1 hour onwards, the level of antioxidants remains unchanged, as a result, the level of free radicals rises to a level higher than normal	(1)

	Concept for mark award	Example	Marks
9(a)	observations from the photographs	the photomicrographs show that the stoma has become smaller / was closed in the presence of plant hormone X	(1)
	inference	a smaller stoma size / closure of stoma will reduce water loss through transpiration, which may help the plant to survive during drought stress / when water supply is inadequate	(1)
9(d)	correct choice	A	(1)
	comparison of data	the leaf of A has lost less water than that of B	(1)
	inference (1)	the leaf of A should have been influenced by hormone X leading to a reduction in size of stomata	(1)

5 marks

DSE
I'esO